ENGINEERS WALLAH

CHEMISTRY

Lecture VI

Mixtures, Its Types and Separation Techniques



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TOPICS *to be covered*

1. Difference Between Mixture and Compound
2. Solutions, Colloids and Suspensions
3. Separation Techniques in One Shot

Knowledge Ride On



Difference Between Mixture and Compound

Knowledge Ride On



Solutions, Colloids and Suspensions

Knowledge Ride On



Separation Techniques in One Shot

Knowledge Ride On



Efficiency Hacks

Glossary

Mixture (मिश्रण)
Compound (यौगिक)
Homogeneous mixture (सजातीय मिश्रण)
Heterogeneous mixture (विजातीय मिश्रण)
Solution (घोल)
Suspension (निलंबन)
Sublimation (ऊर्ध्वपातन)
Distillation (आसवन)
Filtration (निस्पंदन)
Evaporation (वाष्पीकरण)
Centrifugation (केन्द्रापसारण)
Separating funnel (अलग करने की कीप)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Calculate the molecular mass of HNO_3?

RRB JE 2019

$\Rightarrow$ 1 x R.A.M of H + 1 x R.A.M of N + 3 x R.A.M of O

$\Rightarrow$ $1 \times 1 + 1 \times 14 + (3 \times 16)$

$\Rightarrow$ $1 + 14 + 48$

$\Rightarrow$ **63 u**

A 65

B 64

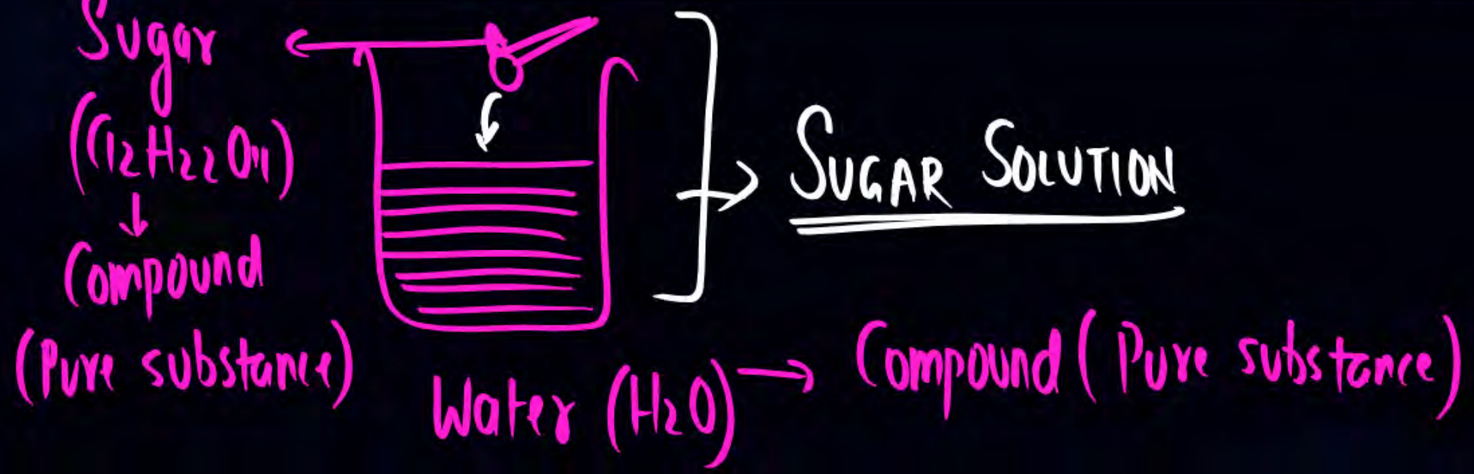
C 70

D 63 ✓

Concept Bites – Difference Between Mixture and Compound

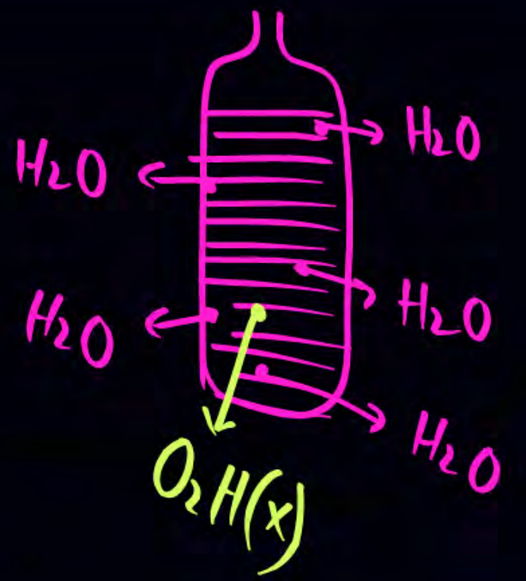
Difference Between Mixture (मिश्रण) and Compound (यौगिक)

Mixture	Compound
A mixture is formed when two or more than two pure substances are mixed together.	A compound is a pure substance formed different elements, i.e. from different atoms.

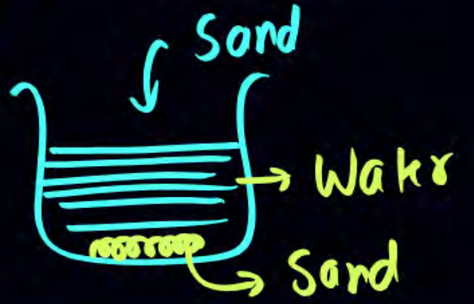


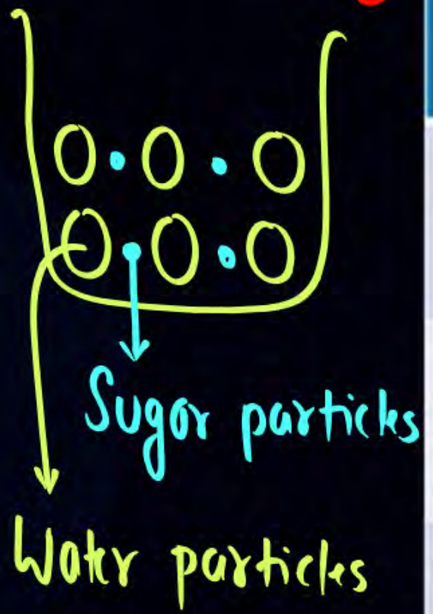
Difference Between Mixture (मिश्रण) and Compound (यौगिक)

Mixture	Compound
A mixture is physically homogeneous or heterogeneous.	A compound is always chemically homogeneous.









Homogeneous Mixture (सजातीय मिश्रण)	Heterogeneous (विजातीय मिश्रण)
(i) Uniform composition (एक समान)	(i) Non-uniform composition एक समान नहीं है।
(ii) They exist in single phase. /single region	(ii) They exist in more than one phase. /region
(iii) There are no visible boundaries of separation.	(iii) There are visible boundaries of separation.

Difference Between Mixture (मिश्रण) and Compound (यौगिक)

Mixture	Compound
A mixture possesses the properties of its individual components. (pure substances)	A compound does not possesses the properties of its individual elements.

Iron filings + Sulphur powder → Mixture

$Fe + S \xrightarrow{heat} FeS$ (Compound)

(I) Bring a magnet near the mixture. ⇒ Iron filings will be attracted

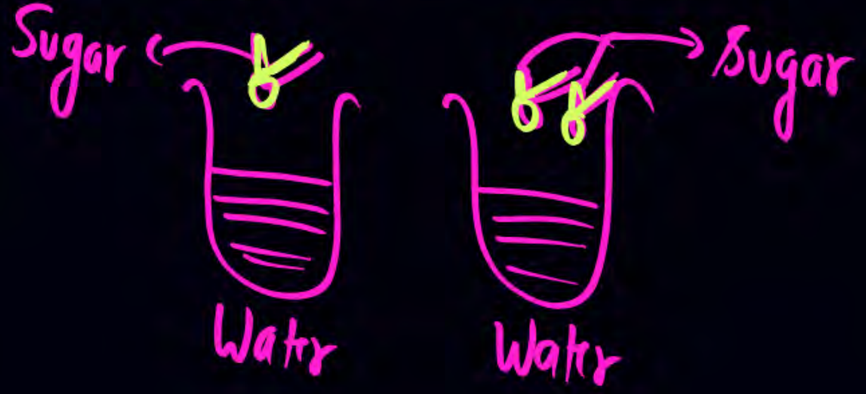
(II) Put the mixture in carbon disulphide (CS_2) ↳ Sulphur will dissolve

Won't show (I) & (II)

Difference Between Mixture (मिश्रण) and Compound (यौगिक)

Mixture	Compound
It does not have fixed melting and boiling points.	It has fixed melting and boiling points.

Variable composition

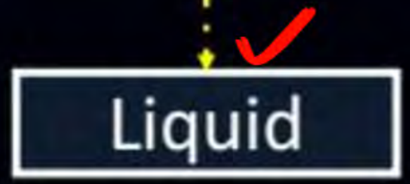


fixed chemical composition

Recap

Physical Classification of Matter







= Plasma

= B.E.C.

Chemical Classification of Matter

- Pure Substance
  - Element
    - Metals
    - Non-metals
    - Metalloids
    - Noble gases
  - Compound
- Impure Substance/mixture
  - Homogeneous
  - Heterogeneous

Practice and PYQ's Wallah



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Question

Which among the following is not true for a mixture?

(false)

A. A mixture has a variable composition. ✓ (TRUE)

B. Mixture don't show the properties of its components. ✗ (FALSE)

C. A mixture does not have fixed melting and boiling points. ✓ (TRUE)

D. A mixture possesses the properties of its components. ✓ (TRUE)

Concept Bites – Solutions, Colloids and Suspensions

1 nm = 10^{-9} m

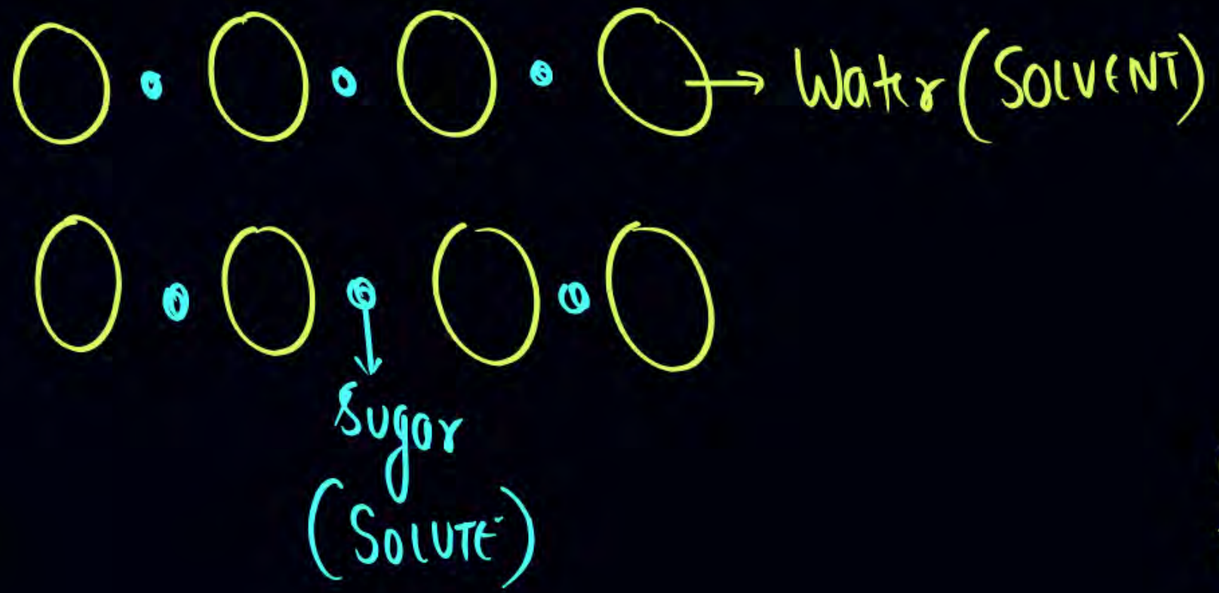
Sugar / Salt in water | Milk, Paint etc. | Oil-in-water, Chalk in water etc.

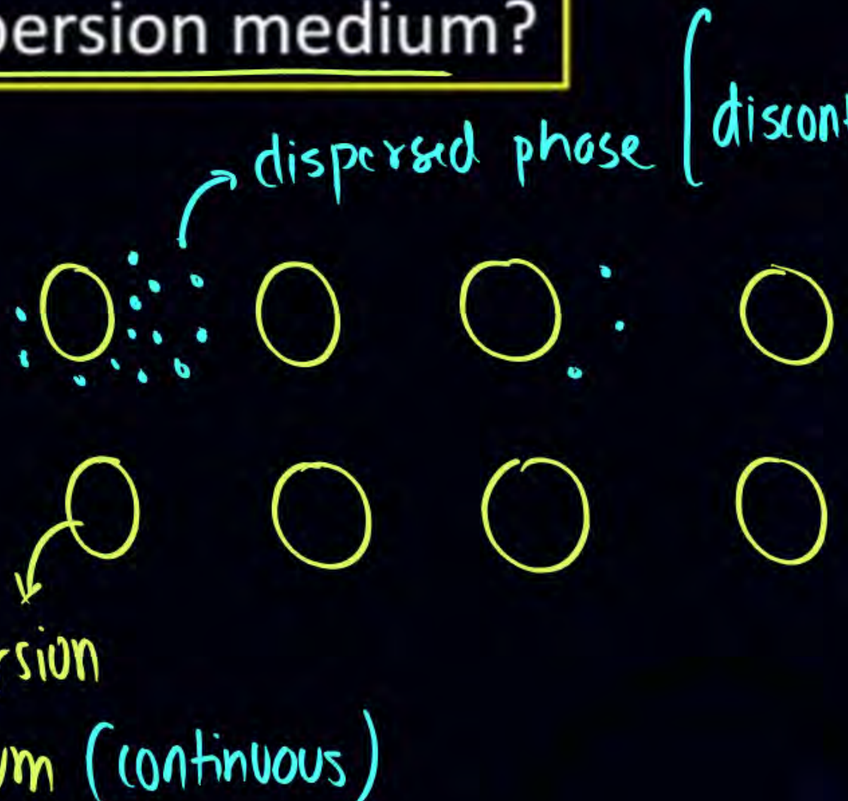
Property (गुण)	Solution (घोल)	Colloidal sol / Colloid	Suspension (निलंबन)
Nature of mixture	Homogeneous	Appears homogeneous but are heterogeneous	Heterogeneous
Particle size	Less than 1 nm	1 nm - 1000 nm	More than 1000 nm
Filtrability (Particles separated by filter paper & semi-permeable membrane)	Filter paper → 'NO' Semi-permeable membrane → 'NO'	'NO' 'YES'	YES YES
Tyndall effect	NO	YES	YES [fine suspension]
Visibility of particles	NO	NO Ultramicroscope: Yes	Yes
Stability (Particles don't settle down) →	YES	YES	NO
Appearance	TRANSPARENT	TRANSLUCENT & OPAQUE	OPAQUE AND MURKY 'Dark'

Try to Answer

Colloids | Colloidal sol

What is a dispersed phase and dispersion medium?





On the Basis of Physical State of Dispersed Phase and Dispersion Medium

Dispersed phase	Dispersion medium	Type	Example
Solid	Solid	Solid sol	Gemstones (रत्न) — Blue sapphire, Ruby etc.
Liquid	Solid	Gel	Cheese, Butter, Curd
Gas	Solid	Solid foam	Pumice stone, Sponge
Solid	Liquid	Sol	Paint, Blood plasma
Liquid	Liquid	Emulsion (Oil-in-water)	Milk, Face cream
Gas	Liquid	Foam	Shaving cream, Soap lather
Solid	Gas	Solid aerosol	→ Smoke, Dust-storm
Liquid	Gas	Aerosol	→ Cloud, Fog, Mist

D.M. → Water
D.P. → fats, protein etc.

Practice and PYQ's Wallah



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Question

Which of the following is a heterogeneous mixture?

RRB JE 2014

A Brass → Alloy of Cu and Zn ⟶ Homogeneous mixture

B Sugar solution in water → Homogeneous

C Air ⟶ Homogeneous

D Milk → Colloid ⟶ heterogeneous

Air → Heterogeneous when dust particles are considered

Question

A gel is a _____.

A liquid dispersed in a solid dispersion medium [Cheese, Curd etc.]

B Solid dispersed in a liquid dispersion medium

C Gas trapped in a viscous liquid

D Gas trapped in a semi-solid

Question

Which among the following is the correct for suspensions?

A. Nature: Homogeneous, Size: Less than 1 nm

B. Nature: Homogeneous, Size: Between 1 nm - 1000 nm

C. Nature: Heterogeneous, Size: Between 1 nm - 1000 nm

D. Nature: Heterogenous, Size: More than 1000 nm

Question

Soap lather is a type of _foam_ in which the dispersed phase is _gas_ and the dispersion medium is _liquid_.

A. Sol, solid, liquid

B. Foam, solid, liquid

C. Foam, gas, liquid

D. Aerosol, liquid, gas

Concept Bites – Separation Techniques in One Shot

Separation Techniques in One Shot

Distillation
(Evaporation followed by condensation)

Type of Mixture	Example	Separation Method	Principle of Separation
Solid-solid mixture	Ammonium chloride (NH_4Cl) + Salt ($NaCl$)	Sublimation (ऊर्ध्वपातन)	One component is sublimable and another non-sublimable
Solid-liquid mixture (Solid component insoluble in liquid)	Sand + Water	Filtration (निस्पंदन)	Insoluble solid won't pass through filter paper while liquid can.
Solid-liquid mixture (Solid component soluble in liquid)	Common salt + Water	Evaporation (वाष्पीकरण)/Distillation (आसवन)	One component is volatile and another non-volatile

sublimable

non-sublimable

does not easily convert to vapour

easily converts to vapour (low b.p.)

Separation Techniques in One Shot

Type of Mixture	Example	Separation Method	Principle of Separation
Solid-liquid mixture (Heterogeneous)	Cream from milk	Centrifugation (केन्द्रापसारण)	Difference in density
Liquid-liquid mixture (Immiscible liquids)	Kerosene oil + Water	Separating funnel (अलग करने की कीप)	Immiscible liquids and form separate layers
Liquid-liquid mixture (Miscible liquids) आपस में घुलने वाले	Acetone + Water 56 °C 100 °C	Distillation (आसवन)	Difference between boiling points more than 25 °C
Liquid-liquid mixture (Miscible liquids)	Ethanol + Water 78 °C 100 °C	Fractional Distillation (आंशिक आसवन)	Difference between boiling points less than 25 °C

आपस में न घुलने वाले

used for colloids

using centrifugal force → forced sedimentation

on the difference in density

Petroleum

→ blood components separation (RBC, WBC, Platelets & Plasma)

↓

Centrifugation

→ Separating different dyes of black ink → Chromatography

→ Drugs identification in bloods of criminals → —11—

Practice and PYQ's Wallah

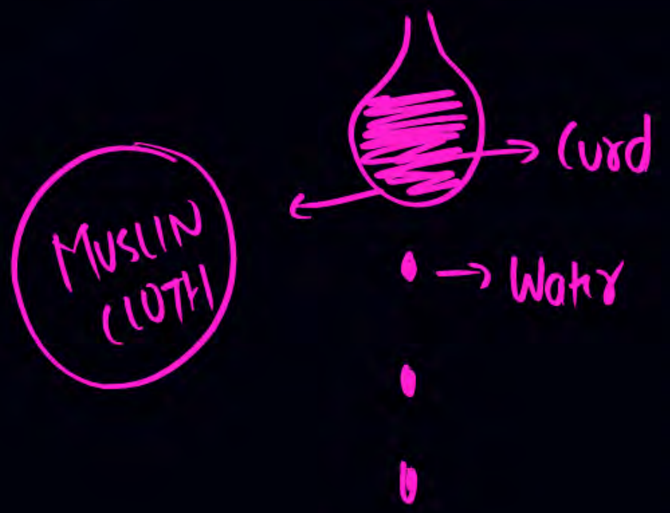


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Question

Which method can be used to separate yoghurt (curd) from water contained in it?

RRB JE 2015



A Distillation

B Crystallisation

C Filtration

D Decantation

Question

Immiscible liquids

Mixture of water and carbon tetrachloride forms separate layers. Which among the following separation technique can separate the two components?

A Distillation

B Handpicking

C Separating funnel

D Sublimation

Question

Which of the following mixtures would you be able to separate using the method of filtration?

A. Oil in water (Immiscible liquids)

B. Cornflakes in milk (Insoluble solid in liquid)

C. Salt in water [Soluble solid in liquid]

D. Sugar in milk [Soluble solid in liquid]

Efficiency Hack!

Tips to Score High

- Exercise (✓)
- Stay hydrated (✓)
- Distractions (✗)
- Healthy Snacks and Foods (✓)



#sbsathhai #pwsathhai

Thank You !